CSA6303 - THREAT INTELLIGENCE AND NETWORK SECUR

Assessment Tool 1: CONCEPT MAPPING

Faculty : Dr.V.Parthipan **SLOT A**

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats | Introduction to Threat Intelligence Types of Threat Intelligence (Strategic, Tactical, perational) — Threat Intelligence Lifecycle — Applications in IT Industry

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

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Code of Conduct:

*I **T. Bhanu Pavan Varma(192425380)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

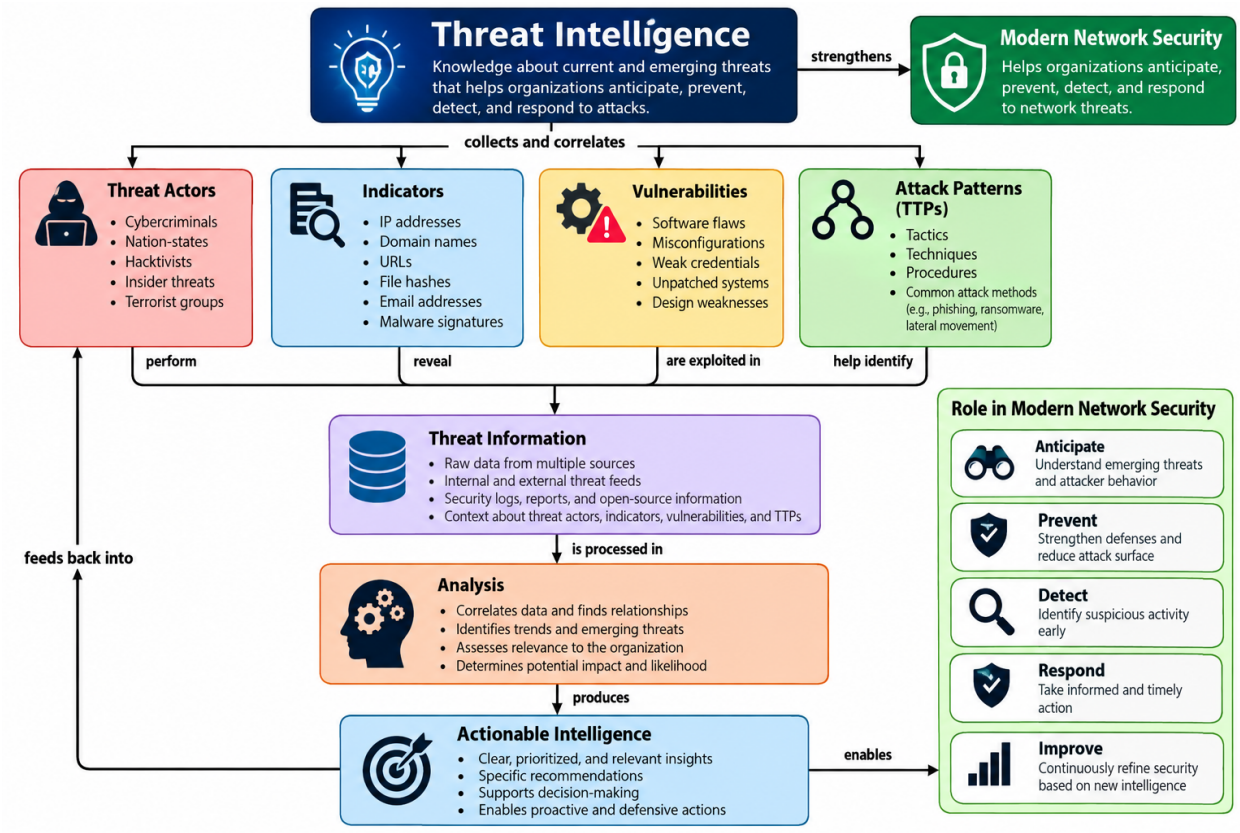
Signature of the student

CO1 – Concept Mapping Questions

No.	Concept-Mapping Question
1	**Construct a comprehensive concept map explaining the foundations of Network Security.** Connect network security fundamentals with security objectives, the **CIA Triad (Confidentiality, Integrity, and Availability)**, network assets, vulnerabilities, threats, and protective measures, and explain how these concepts collectively support secure network operations.
2	**Develop a concept map illustrating the CIA Triad as the foundation of network security.** Relate Confidentiality, Integrity, and Availability to different types of network threats and demonstrate how a single threat can affect one or more security goals.
3	**Construct a concept map classifying different types of Network Threats and their potential impact on security goals.** Connect threats such as malware, phishing, denial-of-service, spoofing, eavesdropping, and unauthorized access with the CIA Triad and explain the resulting security implications.
4	**Develop an integrated concept map connecting Network Security Fundamentals, Network Threats, and Threat Intelligence.** Show how identifying assets, vulnerabilities, and threats leads to the collection and analysis of threat information and ultimately supports informed security decisions.
5	**Construct a concept map explaining the concept of Threat Intelligence and its role in modern network security.** Relate threat actors, indicators, vulnerabilities, attack patterns, threat information, analysis, and actionable intelligence to demonstrate how Threat Intelligence helps organizations anticipate and respond to network threats.
6	**Develop a comparative concept map of Strategic, Tactical, and Operational Threat Intelligence.** Connect each type with its purpose, target audience, time horizon, information requirements, decision-making process, and appropriate network-security use cases, and analyze how the three intelligence types complement one another.
7	**Construct a concept map representing the complete Threat Intelligence Lifecycle.** Connect planning/direction, collection, processing, analysis, dissemination, and feedback with the identification of network threats and security requirements, showing how feedback continuously improves the intelligence process.
8	**Develop an end-to-end concept map showing how a network threat progresses from identification to actionable Threat Intelligence.** Connect network security fundamentals, CIA security goals, threat detection, intelligence collection, processing, analysis, dissemination, and response, and explain the relationship between each stage.

No.	Concept-Mapping Question
9	**Create a scenario-based concept map for an organization experiencing a major network attack.** Integrate the CIA Triad, types of network threats, Strategic Intelligence, Tactical Intelligence, Operational Intelligence, and the Threat Intelligence Lifecycle to demonstrate how the organization can understand, prioritize, and respond to the threat.
10	**Design a comprehensive concept map demonstrating the applications of Network Security and Threat Intelligence in the IT industry.** Connect network security fundamentals, CIA Triad, network threats, Threat Intelligence, Strategic/Tactical/ Operational Intelligence, and the Threat Intelligence Lifecycle with applications such as threat monitoring, incident response, vulnerability management, risk assessment, security operations, and organizational decision-making.

5. **Construct a concept map explaining the concept of Threat Intelligence and its role in modern network security.** Relate threat actors, indicators, vulnerabilities, attack patterns, threat information, analysis, and actionable intelligence to demonstrate how Threat Intelligence helps organizations anticipate and respond to network threats.



7. **Construct a concept map representing the complete Threat Intelligence Lifecycle.** Connect planning/direction, collection, processing, analysis, dissemination, and feedback with the identification of network threats and security requirements, showing how feedback continuously improves the intelligence process.

